CHAPTER 4: Certifying Integrated Pest Management- A Case Study in Defining Eco-Agriculture Strategies

Abstract:

Categorization of farming systems as using or not-using sustainable strategies is essential to contemporary agrifood-systems governance, including green food certifications. Ecologically friendly farming systems are based on adaptive application of principles and tacit knowledge. This makes their categorization more difficult. This paper examines how 22 different farming eco-certifications operationalize the concept of "Integrated Pest Management" (IPM) to classify farms as worthy or unworthy of certification. I qualitatively code the farm inspection checklists based on both a list of 8 IPM principles, and themes that emerged during the coding process. Few certification programs covered all eight principles of IPM, and administrative requirements such as planning, record-keeping and education were more common than requirements relating to specific pest management practices. Compared to other groups, regional single-product certifications were far more detailed in their standards, gave system-specific best management practices and monitoring criteria, and gave more focus to grower collective-action problems such as biosecurity and pest resistance management. Other certifications generally focused on creating a "floor" to mitigate the worst excesses; many certifications focused on Low and Middle-Income Countries were especially focused on issues such as banning the most harmful pesticides, proper disposal of hazardous waste, and worker safety when handling pesticides. These results suggest a duality in certifications- while in well-bounded communities of farmers, certifications may be able to codify frameworks for pursuing an ideal of sustainable farming, broadly applicable certifications may be best suited to minimum standards.

4.1. Introduction

Agriculture plays a critical role in many global social and environmental challenges, including climate change, biodiversity loss, eutrophication, economic development, and food security (Campbell et al., 2017a). To meet these challenges, farmers, scientists, and civil society groups have developed and promoted many new farming systems strategies for improving agriculture agronomically, socially and environmentally. Some of these strategies[10] are associated with grassroots social

[10] In this paper "strategy" refers to any named approach to farming that involves a set of practices, a set of principles, and/or a set of design and decision-making frameworks. Multiple actors may disagree about what "actually" defines a given strategy, it is only important that its promoters see it as a distinctive

movements, such as agroecology, Zero Budget Natural Farming (Khadse & Rosset, 2019), and the System of Rice Intensification. Others, like Climate-Smart and Conservation Agriculture, are more associated with agribusiness and governments. Despite their differences, promoters of many of these strategies conceive them as design or decision-making frameworks based on principles that can be adapted to a wide range of agroecological and social contexts.

If food-systems actors want to promote their preferred strategy through subsidies or certification, or to decide which strategies to promote through research and evaluation, classifying farms as using or not-using these strategies is an important step. This may be simple for some, such as agroforestry, which is defined simply by growing trees alongside other agricultural elements. But it may be much harder for others, where the defining features are principles or frameworks used to design, plan and adapt farming systems to fit within a specific ecological context. Organizations promoting such strategies face difficulty tracking their adoption and may use a simplified definition of a strategy as the presence or absence of a few well-defined practices (Giller et al., 2009). These difficulties also occur in scientific inquiry, where sharp divides can emerge between researchers who study a strategy as a "recipe" (Lyon et al., 2011) through controlled trials and those who study it as a flexible set of principles to be examined through case studies of real-world implementation. These divides are particularly stark regarding "Holistic Management Planned Grazing" (Briske et al., 2008; Gosnell et al., 2020) and the System of Rice Intensification (Glover, 2011; Sheehy et al., 2005).

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approach to farming. Organic agriculture, precision agriculture, management-intensive grazing, permaculture and vertical farming all qualify as strategies under this definition, in addition to those noted in the text.

Certifications for farming systems and products, meant to mark the implementation of a sustainable strategy, are especially reliant on such dichotomous classification. Such certifications have come to play a large role in governing food systems in the "neo-liberal turn" towards non-state governance in the late 20th century (Vogel, 2008). Between 1985 and 2000, most high-income countries developed frameworks for certifying organic agriculture, "Fair Trade" shifted from grassroots relationships into certifications standards and several environmental NGOs launched agriculture, fisheries, and forestry certification schemes. Since that time, numerous other production standards have been developed by corporations seeking to profit from consumer interest in greener foods, NGOs seeking to promote sustainability and producer organizations promoting better practices and product differentiation.

Understanding how certification standards operationalize broad notions of more sustainable agriculture is important for two reasons. First, certification is itself an important component of the modern agri-food system. Second, the profusion of certifications provides an ideal environment for exploring the variety of ways that the same contextual, embedded agricultural strategies can be converted into classification procedure. Such classifications are also important for traditional policy interventions such as mandates, bans or subsidies, and undergird comparative research of farming systems.

This paper examines how sustainability-focused food and agriculture certifications operationalize the concept of Integrated Pest Management. Through an exploratory content analysis, I explore the diversity of ways in which a single concept can be turned into a standard. Furthermore, I analyze trends in how standards with different foci and scopes operationalize the same concepts. These results provide new

insights into what certification and other agricultural systems classifications, try to do and can do, and in what contexts.

4.1.1 Problems in Certification

Ecological food certifications enable consumers to choose products that have un-observable characteristics that they favor, without having a personal relationship with the producer. These characteristics may be (believed to be) beneficial to the consumer, such as lower levels of pesticide residues, or conform to the consumer's ethical and political values, such as environmental stewardship or fairness for small farmers and farmworkers.

Alternative food movements developed ecological and social certification of agricultural products as a tactic to expand their reach. The Demeter Biodynamic Farming standards and trademark were established in 1928 by a German biodynamic producers' cooperative to market produce to consumers (Demter International, n.d.). In the mid-late 20th century, regional associations of organic farmers in the U.S. and Europe developed certification standards to identify foods that were produced on farms following organic principles (Schmid, 2007). In the 1980s and 1990s, environmental and solidarity NGOs developed standards for farming in low and middle income countries (LMIC) to support small farmers and protect biodiversity (Raynolds, 2000). In all these cases, certification was an effort to incorporate consumers and consumption into social movements.

While these certifications have created large alternative markets, provided economic opportunities for small farmers, and encouraged more sustainable practices, many argue that certifications have failed to live up to their potential. Anthropologists have documented perverse impacts from bureaucratic standards, including the

fragmentation of traditional communities and exacerbation of local inequality (Bacon, 2010; Getz & Shreck, 2006b; Mutersbaugh, 2002). A substantial proportion of the price premium for certified products may be consumed by verification and certification costs (Mutersbaugh & Klooster, 2010). As the scale and reach of organic agriculture has grown, more-universalized organic standards have encouraged rather narrow changes in practices and inputs rather than the redesign of farming systems (Darnhofer et al., 2010; Guthman, 2000; P. M. Rosset & Altieri, 1997). Organic standards may still allow for intensification of production at the expense of environmental quality (Mutersbaugh & Klooster, 2010).

Complexity and adaptation are a central problem for alternative food certifications. Early certifications were developed to empower small farmers to continue and deepen their existing ecological stewardship through price premia and technical assistance. Certification, however, necessarily limits the actions that a farmer could take. This represents a problem given the widespread belief that organic (and other ecological) farming is not based on a "recipe" (Lyon et al., 2011) but is rather built on contextual and reciprocal relations with a particular local ecology (Bell et al., 2008; Jackson & Berry, 2009). This makes developing rules to sort farms into or out of a particular view of "ecological agriculture" extremely tricky. In the case of organics, this difficulty drove rule-makers to collapse a complex philosophy into a set of record-keeping requirements and prohibited practices and materials, as discussed in Chapter 3.

Each sustainable farming certifications attempts to achieve multiple goals and faces trade-offs between these goals. Early organic standards were primarily focused on connecting already-existing organic farms with consumers who wanted their goods.

Farmers received recognition and a price premium, consumers gained access to foods produced the way they wanted. Today, organizations promoting organic and other certifications pursue growth in order to expand conversion to sustainable practices and access to certified products. Increasing the number of certified acres or amount of product sold is an important operational objective of certification programs. This creates tensions between realistically achievable and highly stringent standards. The focus on scaling-up may have degraded stringency in Organic and Fair Trade standards (Guthman, 2004; Jaffee, 2012; Jaffee & Howard, 2010). Different standards operating in the same areas adopt different positions regarding this trade-off; in sustainable coffee, Rainforest Alliance and Utz Kapeh deliberately adopted looser, more-attainable standards, to extend the protection of these standards to as many people and farms as possible (Raynolds et al., 2007).

4.1.2 Certification, Boundaries and Integrated Pest Management

Integrated Pest Management (IPM) is one of the most prominent frameworks for better balancing agronomic, social and environmental goals in agricultural systems. First formulated in 1959 (Stern et al., 1959), IPM is now promoted by a wide array of organizations throughout the world as a means to increase or maintain yields while reducing the financial, environmental and social costs of agrochemical use. Like other sustainability-focused farming strategies, IPM is notoriously difficult to define (Bajwa & Kogan, 1996); it is alternately referred to as a "philosophy" (Sappington, 2014), "a way of thinking" (Maupin & Norton, 2010) or a "decision-support system" (Kogan, 1998).

Difficulties in classifying farms as using/not-using IPM have frustrated efforts to measure its adoption and impact (Castle & Naranjo, 2009; Ehler, 2006; Maupin & Norton, 2010; Sappington, 2014; Zalucki et al., 2009).

Using IPM as a basis for certifying more sustainable agriculture poses a difficult problem. IPM's branding as a decision-making framework and a toolkit make it difficult to implement it as a dichotomous classification. As a decision-making framework, IPM is in-principle applicable in nearly any farming system, no matter how biodiverse or input-intensive. Because the decision-making weighs multiple contextual factors, it may not be possible to determine adherence to IPM by merely looking at management decisions, rather than contextually evaluating *why* these decisions were made. Furthermore, as a toolkit of tactics, categorizing farms as "IPM Practitioners" could be extremely inclusive, counting farms that utilize any IPM tactics, or extremely narrow, only including those which use a wide range of tactics. In an extreme example of this difficulty, West and Cisse (2014) found that, among farmers in Quebec, self-identifying as an "IPM Practitioner" was associated with *lower* scores on an index measuring IPM adoption.

Like other sustainability-focused agriculture concepts, IPM's goals may be difficult to incorporate into a certification. The general goal of IPM is to "optimize" or "minimize" the use of pesticides and herbicides- IPM's basic goal is "Spray as little as you possibly can (Castle & Naranjo, 2009)." But minimization can take place within very different sets of constraints: a farmer who sprays to avoid a crop failure and falling into extreme poverty, and a farmer who sprays to avoid cosmetic damage and thus lower prices might both believe that they are spraying "as little as possible." This difficulty shows up prominently in the history of organic agriculture and its move towards

certification. Before certification became ubiquitous, organic farmers strove to minimize agrochemical usage, but a substantial minority still utilized synthetic pesticides, herbicides or fertilizers in a limited fashion (USDA Study Team on Organic Farming, 1980, Chapter 3).Rather than developing a complex framework for including and judging these attempts at minimization within organic certification, organic regulations coalesced behind banning these products entirely.

Despite these difficulties, IPM is often incorporated into sustainable food certifications, including well-know ones such as various "Fair Trade" labels. pesticide use is an important element in agricultural sustainability and health concerns. Consumer's concerns about their exposure to pesticides is a primary driver of organic food purchases, as are concerns about environmental and social impacts of pesticide use (Hughner et al., 2007). Exposure to pesticides is a major occupational risk factor for farm-workers (Tago et al., 2014) and pesticides may increase birth defects, infant mortality and childhood health problems in agricultural communities (Garry et al., 2002; Regidor et al., 2004; C. Taylor A., 2021). Reducing pesticide use thus unites the concerns about food safety, food quality and environmental and social externalities that motivate many people to buy eco-labeled products.

IPM is also an appealing framework because standards can be made more inclusive and attainable than the pest management standards for organic or biodynamic farms. The difficulty in managing insects, and especially weeds without chemicals is frequently cited by farmers as a barrier to converting to organic farming (Khaledi et al., 2011; Stephenson et al., 2017), or as a reason for discontinuing it (Sahm et al., 2013). Standards based on IPM therefore could capitalize on consumer interest in pesticide

reduction while imposing smaller costs on producers, requiring smaller price-premia, and achieving wider market reach.

Because of its broad endorsement, IPM is represented in a wide diversity of certifications. These certifications differ in their geographic scope, from small regions to international, and in their crop scope- from single commodity to any food commodity. Likewise, these programs differ in their origin and intentions, they may come from environmental or development NGOs, university extension services, farmer organizations, food companies or some combination of these. Examining this set of certifications offers insight into the universe of possibilities for certifying and delineating alternative agriculture systems.

4.2 Methods

I assembled a set of certifications which potentially cover foods sold in the United States using the Ecolabel Index (Big Room Inc, 2021), Google search and snowball sampling from websites of certifications found. Inclusion criteria required: that labels are used (directly or indirectly) in the United States, and that the certification protocol references the framework of "Integrated Pest Management." In all, 22 certification programs were identified (Table 4). The certification protocols for these programs were inspected line by line and the criteria related to pest management were coded and categorized. Because some certifications included requirements relating to worker and environmental safety in handling and storage of pesticides under the rubric of IPM, these types of requirements were analyzed for all certifications.

For the purposes of this analysis, "pests" are defined as all organisms that threaten the health and productivity of crop plants; as such, IPM is used as an umbrella term incorporating insect-pest, weed, fungus and disease management. Some certifications adopted this stance, as do some IPM definitions (Kogan, 1998), and for consistency, this was used across all certifications.

Criteria were classified as "Pass-Fail" when the applicant fails certification if the criterion is not met; "Improvement" if the applicant does not have to initially meet the criterion to pass, but after several years in the program, it becomes a requirement; and "Scorecard" if each criterion is assigned a certain number of points, and the applicant must achieve a certain number of total points to pass.

In addition to categories that emerged while coding the sample, criteria were coded based on their relationship to the 8 principles of IPM described by The EU Framework Directive 2009/128/ (OJEC, 2009). This set of principles is similar to those used by many other promoters and is one of the most expansive. The principles are:

1- Prevention and suppression

2- Monitoring

3- Decision-making based on monitoring and thresholds

4- Non-Chemical Methods

5- Pesticide Selection

6- Reduced Pesticide Use

7- Anti-resistance Strategies

8- Evaluation

Criteria could be classified as fitting more than one principle, but criteria were not included if they advanced a principle only through another principle. For example, while pest prevention and reduced pesticide use should also reduce insecticide resistance among targeted pests (because resistance only happens when insecticides are used), criteria focused on prevention or reduced pesticide use are not classified as pertaining to "anti-resistance strategies."

Many certifications included lists of crop protectants whose use is either banned or limited. Many reference one or more lists relating to international treaties or conventions, while others developed their own specific lists of chemicals. All named chemicals were linked to a Chemical Abstracts Service (CAS) Registry Number using a combination of the R package webchem (Szöcs et al., 2020), automated searches of other websites, and manual corrections.

4.2.1 Sample Description

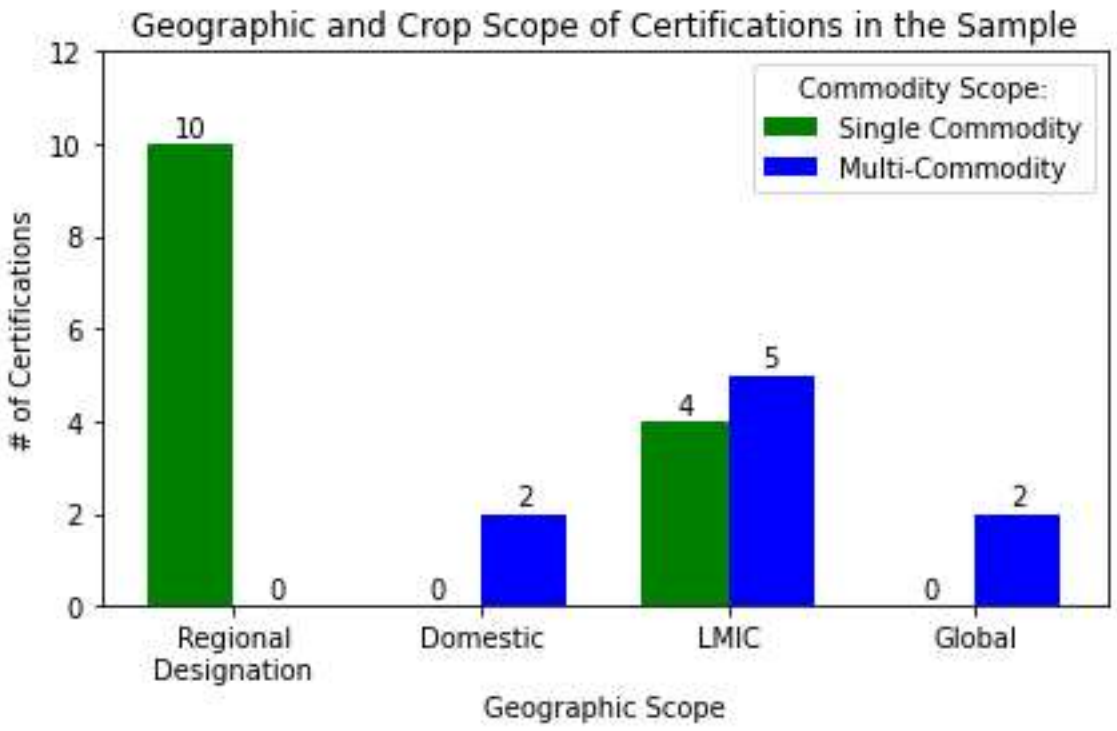


Figure 3: Scopes of Certifications in the Sample

Certifications' scopes were categorized across two dimensions, geographic and commodity group. Certifications had several different geographic scopes, a region within the United States, all areas within the United States, low and middle-income countries (LMIC), or global. Certifications applied either to a single commodity/commodity group or to a wide range of commodities. Figure 3 summarizes the sample categorization.

The composition of the groups which designed and manage the programs differed; seven of 10 regional-designation certifications were developed by grower organizations in partnership with NGOs and/or Universities, while none of the other certifications had a high degree of grower engagement in their development. Three of the single-commodity LMIC certifications were either primarily or jointly developed by

businesses which market the final products. The remaining certifications were developed by environmentally and socially focused NGOs.

Table 4: Sustainable Food Certifications Included in the Study Sample.

Certification Name	Crop Scope	Geographic Scope	Abbreviation
LIVE Certified	Grapes/Wine	Regional	LIVE
Wisconsin Healthy Grown	Potatoes	Regional	HG
Sustainability in Practice	Grapes/Wine	Regional*	SIP
True Earth	Apples	Regional	TE
Bloom Check	Flowers	Regional	BC
Red Tomato Eco Apple	Apples	Regional	EA
California Certified Sustainable Winegrowing	Grapes/Wine	Regional	CCSW
Long Island Sustainable	Grapes/Wine	Regional	LISW
Lodi Rules	Grapes/Wine	Regional*	LODI
Protected Harvest Citrus	Citrus	Regional	P H C
Food Alliance	Many	USA	F A
Sustainable Food Group	Many	USA	S F G
RTRS	Soy	LMIC	RTRS
Better Cotton Initiative	Cotton	LMIC	BCI
RSPO	Palm Oil	LMIC	RSPO
Cotton Made in Africa	Cotton	LMIC (Africa)	CMIA
Fair Trade International	Many	LMIC	FTMH
UTZ	Many	LMIC	UTZ
Fair Trade USA	Many	LMIC	F T USA
Fair For Life	Many	LMIC	F F L
Rainforest Alliance	Many	LMIC	R A
Global GAPS	Many	Global	GGAPS
Farm Sustainability Assessment	Many	Global	FSA

4.3 Results

4.3.1 Certification Structures

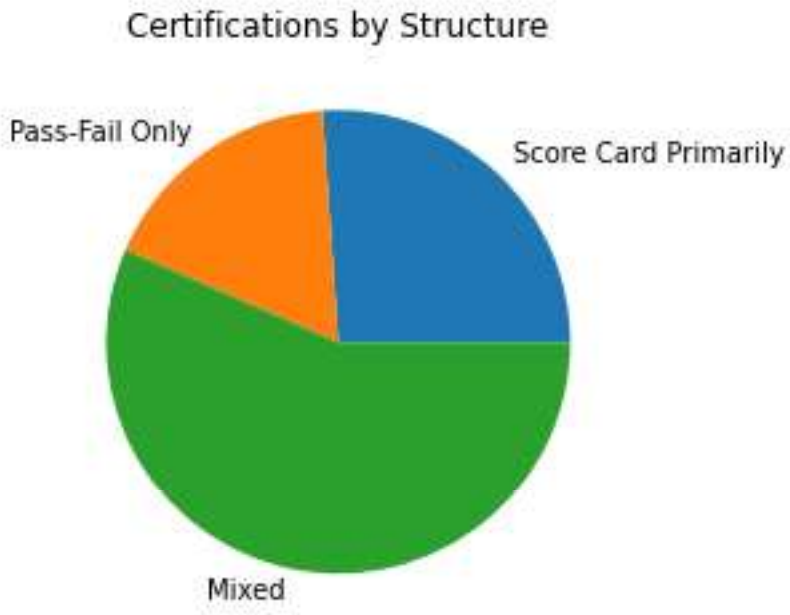


Figure 4: Certifications in the Sample, Grouped by Criteria Structure.

The structure of the certifications is presented in Figure 4, using the categories described above. Six standards were classified as "primarily scorecard" (having three or less pass-fail criteria) with relation to IPM, five were classified as solely pass-fail, and eight were "mixed" between requirements and scorecards and/or improvement.

4.3.1 IPM Principles

Of the eight principles of Integrated Pest Management, two, Monitoring and Pesticide Selection, were mentioned in at least one criterion for all certifications (Figures

5 & 6). Two principles, Thresholds and Evaluation, were mentioned in less than half of standards, though participation in certification itself arguably constitutes a form of evaluation. Resistance Management appeared in only 13 certifications and was referenced more than once in only three standards, but tactics from Principles 1, 3, 4 and 6 all can contribute to managing resistance by reducing pesticide use.

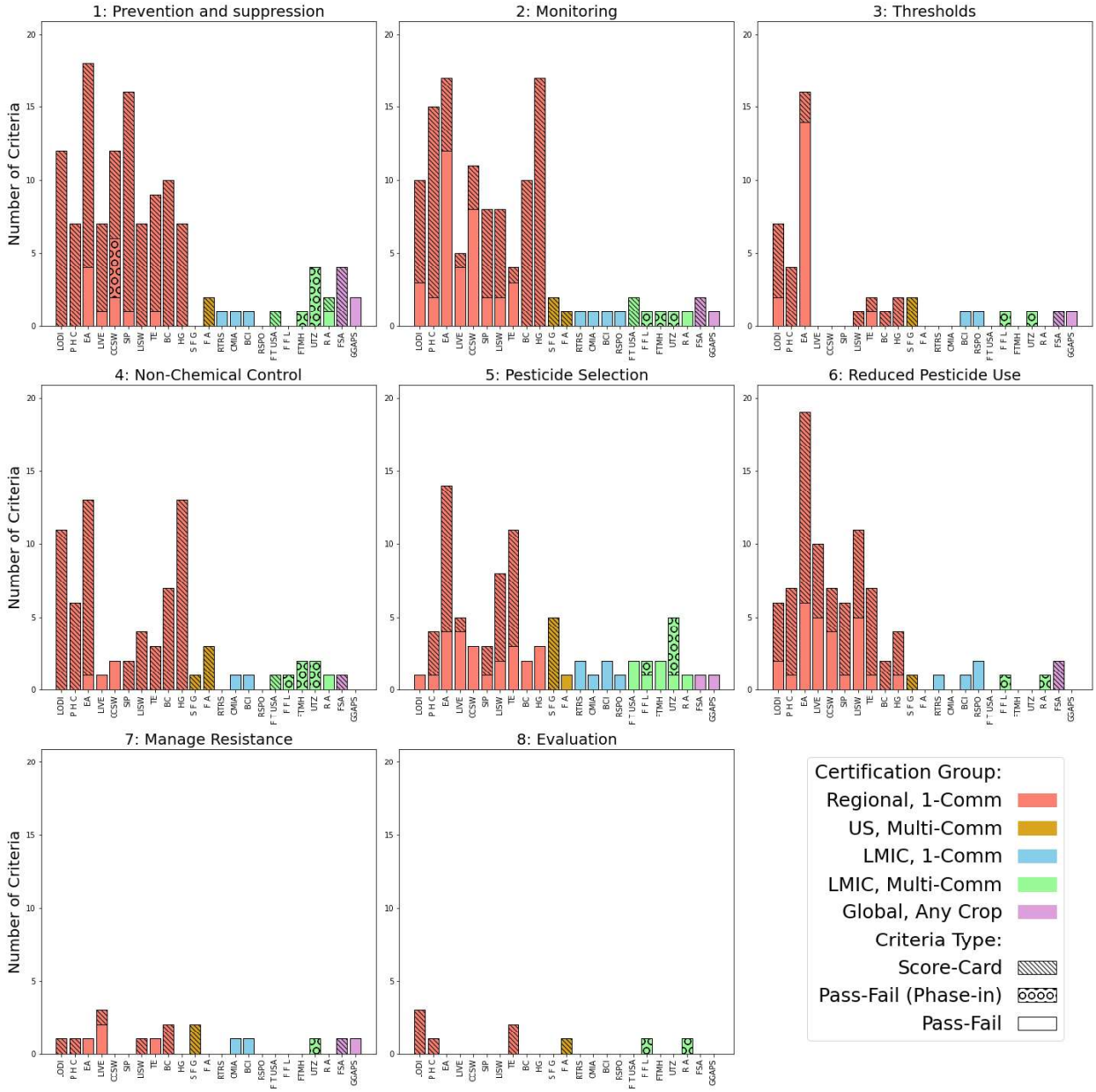


Figure 5: Counts of Criteria in Certifications Pertaining to the 8 Principles of Integrated Pest Management. (Certifications presented in the same order as in Table 4)

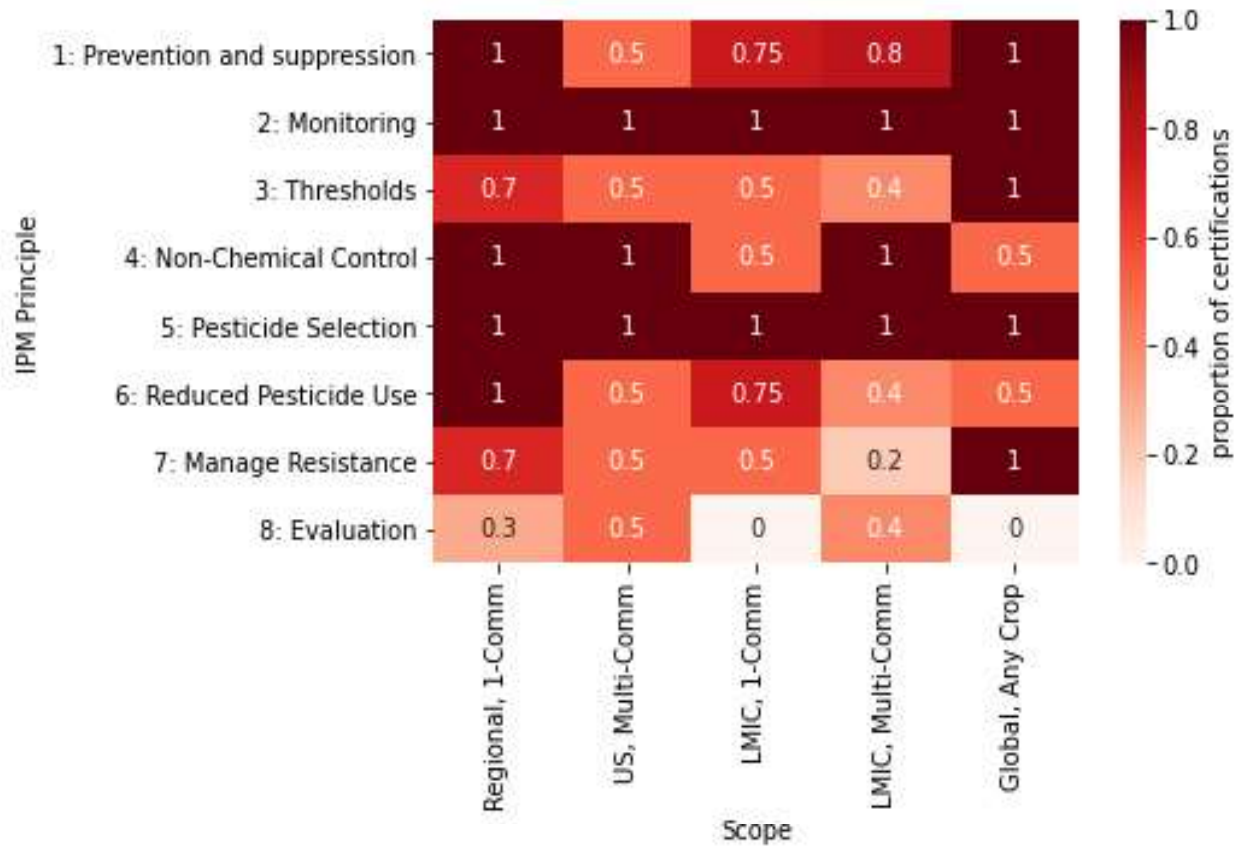


Figure 6: Proportion of Certifications of each scope that have a at least 1 criterion pertaining to each of the eight principles of IPM

4.3.3 Trends Between Groupings

Regional single-commodity certifications differ from other groupings, in their greater detail. These certifications include the most criteria overall. They contain more monitoring, decision-support and action-threshold criteria, often with separate specific protocols for multiple important pests. They are also more likely to promote Mode-of-Action (MOA) rotation for chemical resistance management, and included more criteria relating to biosecurity and sanitation. Further, all certifications which included the use of

specific models for predicting pest pressure or pesticide toxicity models to measure performance were in this group. The difference in number of criteria is stark for most IPM Principles (Figures 5 and 6).

Most regional-designation certifications included multiple region-specific or crop-specific requirements or options for prevention, monitoring and non-chemical control. Nearly all of the other certifications simply mentioned the need to incorporate these areas in a general sense, or to have them mentioned with an IPM plan. Regional-designation certifications also included more criteria for pesticide selection; many include several different environmental considerations, including threats to water quality and beneficial insects. These certifications also include far more criteria for directly reducing pesticide usage, including requirements or options for ensuring correct spray coverage and use of spot-spraying or block-spraying for specific pests.

Certifications focused-on LMIC were different in two respects. First, the number of banned and restricted pesticides tended to be much greater (Figure 7). Many of these pesticides are banned in some countries and not others, a third or more of chemicals banned by many certifications are not registered for any use with the US Environmental Protection Agency.These banned product lists help to create a harmonized global floor. Additionally, these standards place much more emphasis on worker safety and hazardous materials handling. This not only reflects the different missions of some of the certifications, which often foreground social concerns, but also the role that these certifications play in substituting for and supplementing legal standards in agriculture in LMIC.

Like the LMIC-focused certifications, single-commodity regional domestic certifications generally included some worker protections. Most of these commodities, such as grapes, apples and citrus, commonly employ migrant agricultural laborers who may be more vulnerable. Wisconsin Healthy-Grown Potatoes is the only one of these certifications not including any worker protection criteria; migrant farmworkers do not commonly work in this sector.

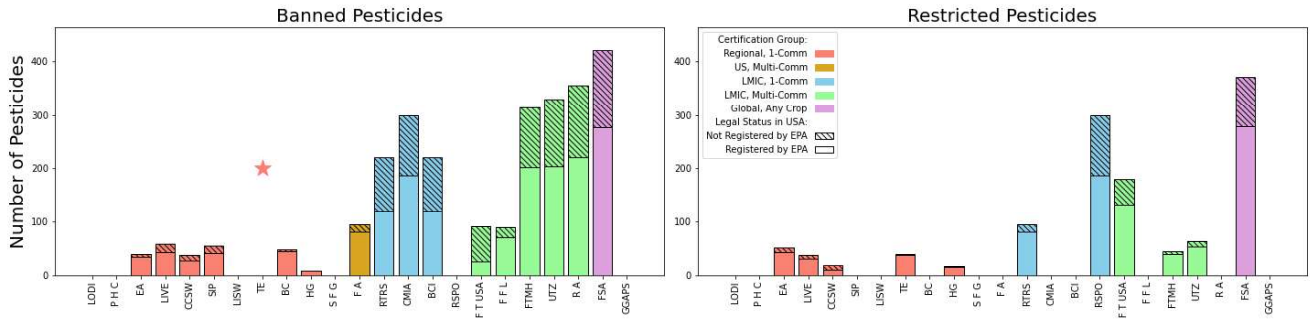


Figure 7: Number of Unique Pesticides (by CAS number) Banned or Restricted by Certifications in the Sample. (Certifications presented in the same order as in Table 4). *True-Earth Certified keeps a short list of 'allowable products'.

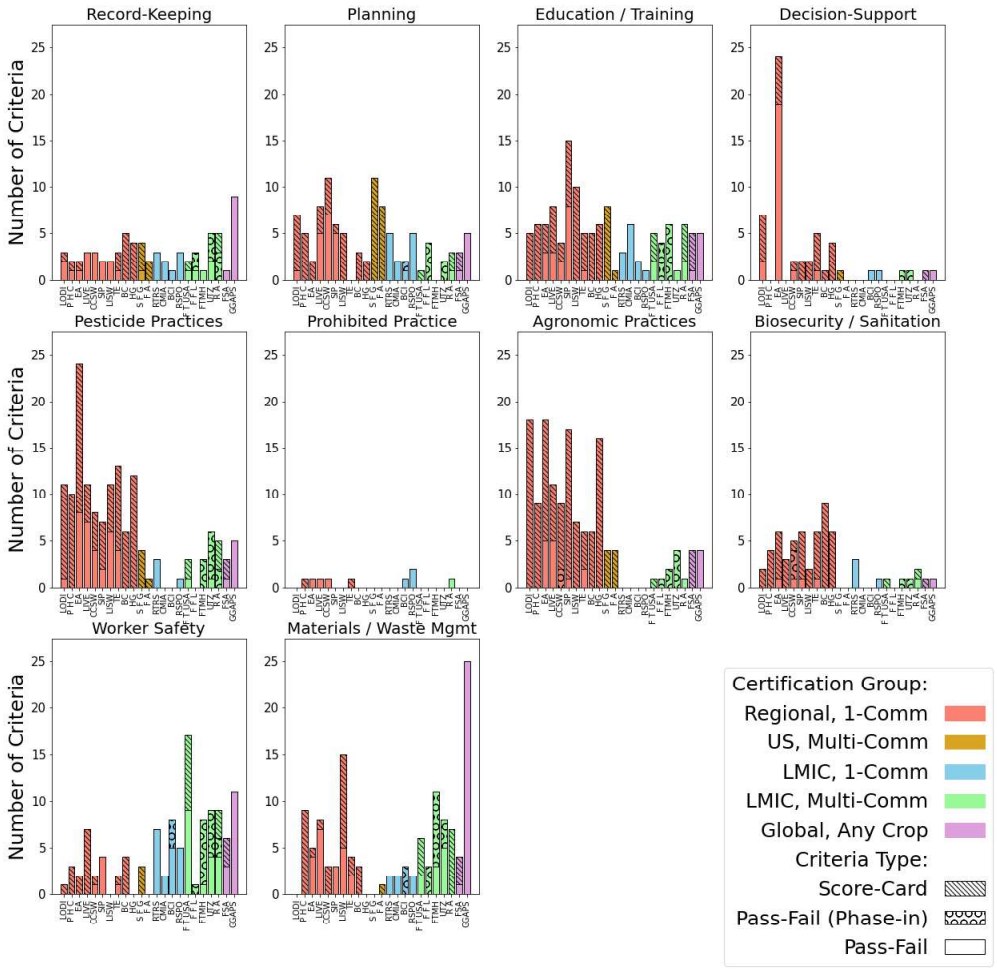


Figure 8: Counts of Criteria in Certification relating to self-coded themes. (Certifications presented in the same order as in Table 4)

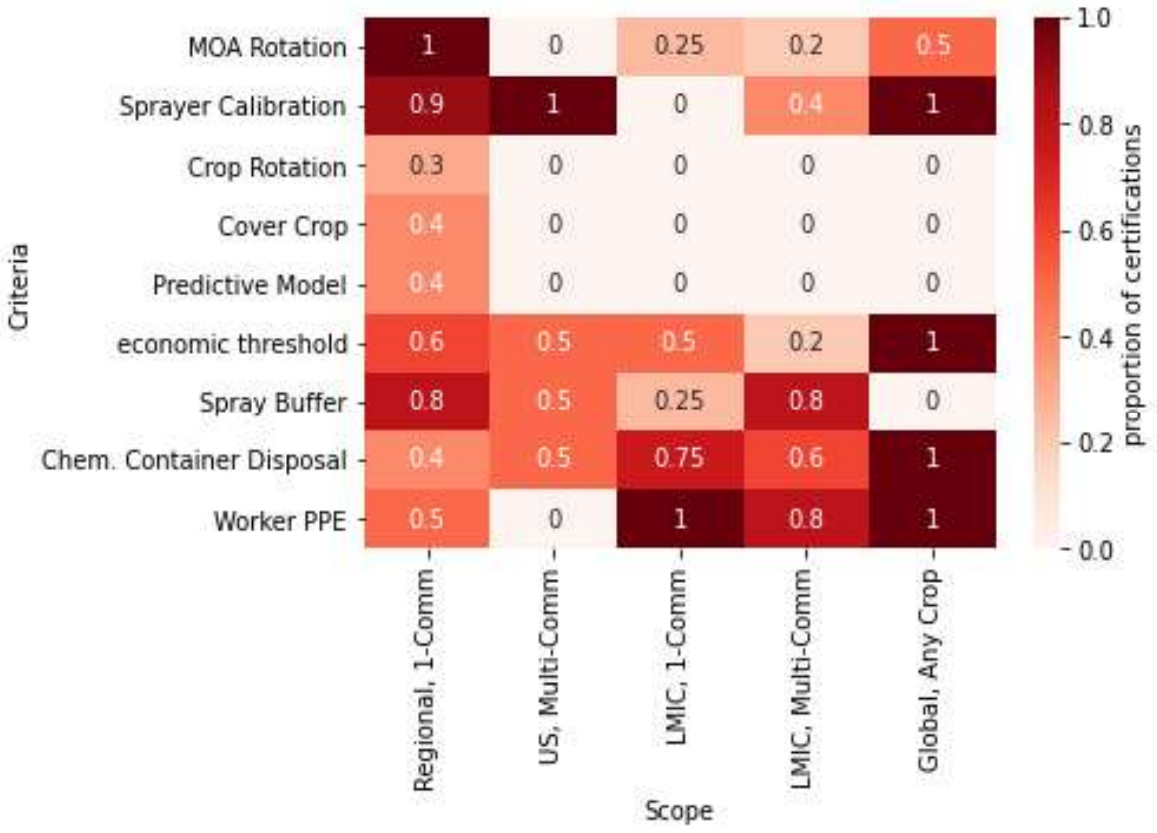


Figure 9: Proportion of Certifications with Different Scopes that have at least 1 Criterion pertaining to sub-themes.

4.4 Discussion

The differences between regional single-commodity standards and all others reflect tension between an "inside" and "outside" view of ecological best management practices in agriculture. Standards with a broad scope must adopt an external view of farming systems- the stringency of a certification is tightly linked to how much it limits farmers' choices. Such standards may be very useful for setting a "floor" which excludes

the worst practices or materials from production but cannot define “best practices” in a rigorous way without ignoring enormous agroecological variation.

Regional-Designation certifications function very differently than others; many appear to incorporate a “view from within” of agricultural best management practices. These differences likely reflect two factors. First, standards for a specific crop in a specific agroecological zone can be more exhaustive and specific. Several of the standards in this group contained monitoring and/or threshold standards for several different insects and diseases. Comprehensive description of monitoring and threshold techniques for “any pest, any crop, anywhere” is obviously not possible within a single guidance document. Similarly, the thresholds for an acceptable toxicity load in a toxicity model may be very different across different crops or agroecological zones.

Additionally, these differences in standards reflect the collective-action and public-goods aspects of IPM. In contrast to many other environmental problems in agriculture, chemical resistance in pests and the import of exotic pests and diseases are public goods problems that are spatially bounded and primarily impact the farmers themselves. The same applies promoting habitat and natural enemies, though these public benefits may be more spatially localized. These issues closely resemble the types of problems successfully dealt with by informal “commons-type’ governance structures (Ostrom et al., 1999). Individual efforts in these areas can have high positive externalities within the community of crop-producers, creating the potential for free-rider problems, but the impacted community is well-bounded. As such, certification provides the opportunity not just for farmers to differentiate their products to consumers, but to agree

to manage key public goods and bads. Resolving such collective-action problems is a well-known key challenge and opportunity in IPM promotion (Bottrell & Schoenly, 2018; Parsa et al., 2014).

Relatedly, these certifications tend to be developed more closely with growers and codify regional best-management practices while allowing growers flexibility. This allows for certification standards to be potentially more rigorous without being overly constraining upon farmers' decision-making. On the other hand, multiple competing regional single commodity certifications may cause consumer confusion; the sample includes three different sustainable winegrowing certifications in California alone. A profusion of certifications may result in advantages to programs with weaker standards. In the case of forestry, Forest Stewardship Council is widely seen as having stricter environmental standards than the industry-driven Sustainable Forestry Initiative, but has failed to have the same reach (Clark & Kozar, 2011).

Globally focused standards face the tradeoff between stringency and flexibility more starkly. The original UTZ and Rainforest Alliance standards were explicitly designed to be low-cost and accessible, extending protection from the worst environmental and social abuses to as many farms as possible (Auld, 2010). The pest management content of these standards and other global-scale standards focuses on worker safety, hazardous materials handling, prohibition of highly toxic pesticides, and general planning and record-keeping. Preventing pests and disease, and cultural/ecological methods for managing pest populations are only mentioned in general terms.

In this manner, standards with broad applicability may follow the template of the organic standard, mostly focusing on practices to be banned. Prescriptions for sustainability will be too contextual to fit into a global standard. Becoming too strict in standards may not only reduce the economic viability of a certification, but also have perverse impacts if a banned tool or technique is occasionally the most ecological friendly option.[11] Requiring planning, record-keeping, monitoring and training may help farmers find opportunities to reduce their impacts, but is unlikely to resolve any inherent conflicts between productivity and environmental concerns.

4.5 Conclusion

I analyzed twenty-two alternative food and agriculture certifications for how they operationalize the concept of "Integrated Pest Management", through their standards. These standards showed markedly different patterns based on their geographic and crop scopes. Standards developed in one region for one crop sometimes function as compendia of best practices with a high degree of specificity. Other standards tend towards enforcing a "floor" of banned materials and practices while also requiring basic planning and record-keeping, which may assist farmers to improve their practices and impacts.

These results emphasize the difficulties in delineating sustainability-focused agricultural strategies rigorously and repeatably. While this study focuses on non-governmental labeling programs, these same problems can arise in any effort to dichotomously categorize farming systems. These results show that such classifications most easily function to uphold basic standards at a broad scale or to collaboratively codify a menu of well-known BMPs on a local/regional scale. Universal checklists for

[11] For instance, organic agriculture's ban on herbicides may promote unsustainable levels of tillage in some farming systems (Carr, 2017), while bans on synthetic pesticides may not result in lower total environmental impact from pest control in some crops (Kovach et al., 1992).

classifying farming systems across different agroecological contexts as using “Integrated Pest Management” or other principles-based strategies are likely not feasible.